$$= \frac{1}{3D^2+D-14}(4)+\frac{1}{3D^2+D-14}\left(-13e^{\frac{-7}{3}x}\right)$$

$$= PI_1 + PI_2$$

$$\text{P.I}_1 = \frac{1}{3D^2+D-14}4e^{0x}$$

$$= \frac{1}{0+0-14}4e^{0x}$$

(replace D *by* 0)

$$\text{P.I}_1 = \frac{-4}{14} = \frac{-2}{7}$$

$$P.I_2 = \frac{1}{3D^2+D-14}\times(-13)e^{\frac{-7}{3}x}$$

replace D by $\frac{-7}{3}$

Here $3D^2+D-14=0$, when $D=-\frac{7}{3}$

$$\therefore\ P.I_2 = x.\frac{1}{6D+1}\left(-13e^{\frac{-7}{3}x}\right)$$

replace D by $\frac{-7}{3}$

$$\therefore\ P.I_2 = x\frac{1}{6\left(\frac{-7}{3}\right)+1}\left(-13e^{\frac{-7}{3}x}\right)$$

$$= x\frac{1}{-13}\left(-13e^{\frac{-7}{3}x}\right)$$

$$= xe^{\frac{-7}{3}x}$$

The general solution is $y = C.F. + P.I_1 + P.I_2$

$$y = Ae^{\frac{-7}{3}x} + Be^{2x} - \frac{2}{7} + xe^{\frac{-7}{3}x}$$

Example 4.32

Suppose that the quantity demanded $Q_d = 29-2p-5\frac{dp}{dt}+\frac{d^2p}{dt^2}$ and quantity supplied $Q_s = 5+4p$ where p is the price. Find the equilibrium price for market clearance.

Solution:

For market clearance, the required condition is $Q_d = Q_s$

$$\Rightarrow 29-2p-5\frac{dp}{dt}+\frac{d^2p}{dt} = 5+4p$$

$$\Rightarrow 24-6p-5\frac{dp}{dt}+\frac{d^2p}{dt^2} = 0$$

$$\Rightarrow \frac{d^2p}{dt^2}-5\frac{dp}{dt}-6p = -24$$

$$(D^2-5D-6)p = -24$$

The auxiliary equation is

$$m^2-5m-6 = 0$$

$$(m-6)(m+1) = 0$$

$$\Rightarrow m = 6, -1$$

$$\text{C.F} = Ae^{6t}+Be^{-t}$$

$$\text{P.I} = \frac{1}{\phi(D)}f(x)$$

$$= \frac{1}{D^2-5D-6}(-24)e^{0t}$$

$$= \frac{-24}{-6}$$ (Replace D by 0)

$$= 4$$

The general solution is $p = C.F + P.I$

$$= Ae^{6t}+Be^{-t}+4$$

Exercise 4.5

Solve the following differential equations

(1) $\frac{d^2y}{dx^2}-6\frac{dy}{dx}+8y=0$

(2) $\frac{d^2y}{dx^2}-4\frac{dy}{dx}+4y=0$

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